

Robust Numerical Parameter Estimation for Photovoltaic Cells

Deterministic Fitting, Safeguarded Current Solving, and Noise Robustness

CSE 402 Numerical Methods Project

Group Number: 11 Section: A

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Abstract

Photovoltaic parameter extraction is a nested nonlinear problem: model parameters must be fitted from measured current-voltage data, while every model evaluation itself requires the solution of an implicit current equation. This project develops a deterministic and auditable pipeline for the five-parameter single-diode model (SDM) and seven-parameter double-diode model (DDM). The outer fit uses scaled Gauss-Newton or Levenberg-Marquardt (LM); the inner current calculation uses Newton, bisection, or a safeguarded Newton-bisection hybrid with automatic bracketing and warm starts. Finite-difference Jacobians, bounded updates, partial-pivot Gaussian elimination, explicit failure propagation, and reproducible Monte Carlo experiments complete the implementation. On the 26-point French RTC cell dataset, the representative LM-hybrid fits attain RMSEs of 7.54005×10^{-4} A for SDM and 6.72893×10^{-4} A for DDM, a 10.76% reduction for DDM. However, the DDM scaled Jacobian is about 171 times more ill-conditioned, and its noise trials show more convergence failures and substantially larger parameter variation. In 800 paired noisy-data fits, SDM converges in all 400 trials, whereas DDM success decreases from 100/100 at 0.1% noise to 83/100 at 2%. The results show that a lower curve-fitting error does not imply reliable physical parameter identification; LM damping and failure-aware reporting are essential for interpreting the more flexible DDM.

Keywords: photovoltaic modeling, parameter estimation, Gauss-Newton, Levenberg-Marquardt, hybrid root finding, Monte Carlo analysis

Code Repository

https://github.com/sadrul18/CSE_402_PROJECT

1 Introduction and Base Paper Review

Photovoltaic (PV) equivalent-circuit models represent measured current-voltage (I-V) behavior with physically meaningful parameters such as photocurrent, diode saturation current, series resistance, shunt resistance, and diode ideality factor. Reliable extraction of these parameters supports simulation, maximum-power-point studies, diagnosis, and controller design. The

difficulty is that current appears both linearly and inside an exponential. Parameter estimation therefore contains an inner root problem inside an outer nonlinear least-squares problem.

The selected base paper by Qin et al. [1] studies both SDM and DDM parameter extraction for the French RTC cell and the Photowatt PWP 201 module. It combines five metaheuristic optimizers (AEO, GBO, GNDO, BO, and RTH) with three current-calculation strategies: approximation, Lambert W, and Newton-Raphson (NR). The paper concludes that Lambert W and NR give nearly identical and more accurate currents than the approximation, while NR uses fewer computational resources. For the RTC cell, its best reported RMSEs are 7.72986×10^{-4} A for SDM and 6.93709×10^{-4} A for DDM.

Our objective is complementary. Instead of reproducing the five metaheuristic searches, we investigate the numerical behavior of a deterministic end-to-end pipeline. The work asks: (i) can scaled Gauss-Newton and LM fit SDM/DDM parameters accurately; (ii) can a safeguarded Newton-bisection solver preserve speed while reducing root failures; and (iii) how do initialization, noise, and conditioning affect convergence and parameter identifiability? The French RTC measurements originate from a long-established benchmark [2] and are taken here from the base paper's supplementary Table S1.

2 Methodology

2.1 Implicit PV Models and Objective

For voltage V , current I , thermal voltage $V_t = kT/q$, and N_s series cells, define $u = V + IR_s$. The SDM root equation is

$$f(I) = I_{ph} - I_0 \exp\left(\frac{u}{nN_sV_t}\right) - \frac{u}{R_{sh}} - I = 0. \quad (1)$$

The DDM adds a second diode branch:

$$f(I) = I_{ph} - I_{01} \exp\left(\frac{u}{n_1N_sV_t}\right) - I_{02} \exp\left(\frac{u}{n_2N_sV_t}\right) - \frac{u}{R_{sh}} - I = 0. \quad (2)$$

The parameter vectors contain five and seven values, respectively. At each measurement i , the inner solver obtains $I_i^{pred}(\theta)$,

and the outer residual is

$$r_i(\theta) = I_i^{pred}(\theta) - I_i^{meas}, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N r_i^2}. \quad (3)$$

2.2 Inner Current Solvers

Newton uses the analytic slope $\partial f / \partial I$ and updates $I_{k+1} = I_k - f(I_k) / f'(I_k)$. It is fast near a root but has no bracket. Bisection first requires an interval $[a, b]$ with $f(a)f(b) \leq 0$ and always selects the midpoint; it is conservative but linearly convergent. The hybrid keeps the sign-changing bracket, attempts a Newton step, and uses it only when the proposal is finite, lies strictly inside the bracket, and does not worsen $|f|$. Otherwise, it immediately falls back to the midpoint. For bisection and hybrid, the code expands a bracket around the current guess for at most 30 attempts.

Voltages are solved in sorted order and the preceding current warm-starts the next point. Results are returned in the original row order. A failed scalar root remains explicitly marked; the residual routine rejects the entire curve rather than dropping the row or using a NaN value.

2.3 Jacobian and Outer Parameter Updates

The residual Jacobian $J_{ij} = \partial r_i / \partial \theta_j$ is computed by bounded finite differences. The nominal perturbation is

$$h_j = 10^{-6} \max(|\theta_j|, s_j), \quad (4)$$

where s_j is a parameter scale. Central differences are used in the interior and one-sided differences at a bound. Scaling is necessary because saturation currents can be near 10^{-7} A while R_{sh} is tens of ohms.

With $J_s = J \text{diag}(s)$, Gauss-Newton solves

$$(J_s^T J_s) \Delta z = -J_s^T r. \quad (5)$$

It tries the largest feasible update and halves the step when a candidate is invalid or non-improving. LM [3, 4] instead solves

$$(J_s^T J_s + \lambda I) \Delta z = -J_s^T r. \quad (6)$$

The implementation starts at $\lambda = 10^{-3}$, multiplies it by 10 after a failed or non-improving proposal, and divides it by 3 after acceptance. Both methods enforce physical bounds and resolve on free coordinates when an active bound blocks motion. The linear systems are solved by custom Gaussian elimination with partial pivoting. This deterministic local strategy follows standard least-squares principles [5].

2.4 Failure and Stopping Policy

The fitter accepts only a complete finite curve with lower mean squared error. If the initial curve fails, no residual or RMSE is reported. A failed candidate triggers Gauss-Newton backtracking or stronger LM damping, for at most 20 trial steps. A Jacobian failure or exhausted trial sequence returns the last accepted iterate with `converged=False`. Successful stopping requires a small gradient, a small scaled parameter change, or a small relative MSE improvement. This distinction prevents a low retained RMSE from being mislabeled as formal convergence.

Table 1: Common experiment configuration.

Item	Setting
Dataset	French RTC cell, 26 I-V points
Temperature / irradiance	306.15 K / 1000 W m ⁻²
Series cells	$N_s = 1$
Inner tolerance / limit	10^{-8} / 100 iterations
Outer tolerance / limit	10^{-8} / 500 iterations
Default method pair	LM + warm hybrid
Noise levels	0.1, 0.5, 1, and 2%
Noise trials	100 per level and model
Initial-value trials	30 per method combination
Random seeds	42–45

3 Implementation and Experiments

3.1 Software Architecture

The implementation uses Python and NumPy [6]; Matplotlib produces the figures [7]. The full source, test suite, and generated report assets are available in the project repository at https://github.com/sadrul18/CSE_402_PROJECT. The call chain is `run.py` $\rightarrow$ `fit.py` $\rightarrow$ `numerics.py` $\rightarrow$ `current.py` $\rightarrow$ `model.py`. The model layer validates inputs and evaluates Eqs. (1)–(2); the current layer implements all three root solvers; the numerical layer forms residuals and Jacobians and solves linear systems; the fitting layer implements GN/LM; and the experiment layer runs comparisons, Monte Carlo trials, exports, and plots. The automated suite contains 52 tests covering equations, derivatives, root failures, bounded Jacobians, optimizers, repeatability, and full-pipeline behavior.

3.2 Data, Bounds, and Protocols

All 26 RTC rows, including negative-voltage and negative-current endpoints, are preserved. The model-specific initial vectors are

$$\begin{aligned} \theta_0^{SDM} &= [\max(I), 5 \times 10^{-7}, 0.1, 50, 1.5], \\ \theta_0^{DDM} &= [\max(I), 3 \times 10^{-7}, 0.05, 60, 1.5, 10^{-4}, 3]. \end{aligned}$$

DDM bounds follow Qin et al.’s Table 1 after converting saturation currents from microamperes to amperes: $I_{ph} \in [0, 1]$, $I_{01}, I_{02} \in [10^{-15}, 10^{-3}]$ A, $R_s \in [0, 0.5]$ Ω , $R_{sh} \in [0.001, 100]$ Ω , $n_1 \in [0.5, 5]$, and $n_2 \in [1, 5]$.

Three experiment families are used. First, all 12 combinations of two models, two outer optimizers, and three inner solvers share the same model-specific starting vector. Second, 30 seeded lognormal perturbations of each start test local initialization robustness. Third, paired Monte Carlo trials add independent Gaussian current noise with $\sigma_i = |I_i|p/100$; SDM and DDM receive the same noisy arrays at each level. Failed fits are retained in failure statistics, while parameter means and standard deviations use converged fits only.

4 Results and Discussion

4.1 Clean-Data Accuracy and Computational Work

Table 2 reports the complete clean comparison. Newton is the least expensive solver on this well-behaved dataset. The hybrid needs more equation work but retains bracket safeguards; bisection is much more expensive and its small solver-level changes can disrupt finite-difference outer steps. The GN-bisection combinations retain good curves but do not meet the optimizer’s stopping contract.

The representative LM-hybrid DDM reduces clean RMSE by 10.76% relative to SDM. Its fitted $n_2 = 5$ lies on the upper bound, so it should be read as a constrained optimum rather than uniquely recovered physics. Figure 1 shows that all successful project curves reproduce the plateau, knee, and high-voltage collapse; differences that are important numerically are nearly invisible at the full I-V/P-V scale.

Absolute-error plots are more discriminating (Fig. 2). For the rounded GNDO reference parameters, NR has lower absolute current error than the paper’s approximation at 19 of 26 voltages. The representative project LM-hybrid DDM has maximum absolute current error 1.182409×10^{-3} A and maximum power error 6.66051×10^{-4} W, both at 0.5633 V.

4.2 Comparison with the Base Paper

The unchanged repository data produce slightly lower RMSE than Qin et al.’s published values. This is not attributed to optimizer superiority. At $V = 0.5833$ V the stored current is -0.1233 A, whereas another common RTC table uses -0.1230 A. A separately labeled in-memory diagnostic changes only that point and nearly recovers the paper levels (Table 3). The main dataset remains unchanged.

Table 3. RMSE provenance check (A).

Source	SDM	DDM
Project data	0.000754005	0.000672893
One-point diagnostic	0.000773006	0.000693726
Qin et al. best NR	0.000772986	0.000693709

The comparison is therefore qualitative and protocol-aware. The base paper uses population-based metaheuristics with 50 agents and 3000 iterations; our outer iteration is a Jacobian-based local update with very different work. We compare terminal RMSE and curve behavior, but do not claim a same-hardware speedup or reconstruct unavailable paper histories.

4.3 Initialization and Convergence Robustness

Figure 3 averages 30 runs per method combination and carries the last retained value forward after stopping. For SDM, GN/LM with Newton or hybrid succeed in all 30 runs. For DDM, LM with Newton or hybrid also succeeds 30/30, whereas GN with either method fails 8/30; a few poor retained fits raise the all-run mean to 0.0646561 A. DDM GN-bisection succeeds only 2/30. Thus, the nine-iteration clean GN result is fast but not robust evidence; LM damping is especially important in the seven-parameter problem.

4.4 Monte Carlo Noise Robustness

The paired study contains 800 fits. Table 4 separates noisy-data fit error from error against the original measurements. DDM fits successful noisy trials slightly more closely at every level, but at 0.5–2% noise it has slightly higher mean clean-reference error and 12–17% stopping failures. SDM converges in all 400 trials under the shared 500-iteration budget.

Parameter variability reveals a stronger limitation than curve RMSE. At 2% noise, SDM I_{ph} has only about 0.71% coefficient of variation (CV), but I_0 and R_{sh} reach 96.31% and 53.53%. DDM is less identifiable: I_{01} reaches 157.62% CV and I_{02} reaches 233.51%, even though its predicted curves remain close. Figure 4 visualizes this separation between stable output curves and unstable physical parameter estimates.

4.5 Conditioning and Identifiability

Table 5. Local conditioning at the LM-hybrid clean fit.

Model	$\kappa(J_s)$	$\kappa(J_s^T J_s)$
SDM	1,110.75	1.2338×10^6
DDM	190,393.87	3.6250×10^{10}

Table 5 explains the convergence and noise behavior. The DDM scaled Jacobian is about 171 times more ill-conditioned than the SDM Jacobian, and the normal equations square the condition number. These are local, scale-dependent sensitivity diagnostics, not covariance estimates or proof of a global rank defect. Nevertheless, their agreement with the initialization failures and high diode-current CVs is strong evidence that DDM curve accuracy does not imply unique parameter recovery.

Table 2: Clean RTC fits. Times are sequential measurements on one machine and should only be compared within this experiment. “Res.” is the number of full residual-vector evaluations.

Model	Outer	Inner	Converged	Iterations	Time (s)	Res.	RMSE (A)
SDM	GN	Newton	yes	12	0.169	151	0.000754004951
SDM	GN	Bisection	no	31	2.427	458	0.000754371549
SDM	GN	Hybrid	yes	12	0.275	151	0.000754004951
SDM	LM	Newton	yes	26	0.315	322	0.000754004951
SDM	LM	Bisection	yes	31	2.121	394	0.000754150866
SDM	LM	Hybrid	yes	26	0.595	322	0.000754004951
DDM	GN	Newton	yes	9	0.160	142	0.000672892705
DDM	GN	Bisection	no	42	5.387	902	0.000679942695
DDM	GN	Hybrid	yes	9	0.294	142	0.000672892705
DDM	LM	Newton	yes	120	2.229	1965	0.000672892705
DDM	LM	Bisection	yes	75	7.344	1243	0.000689650917
DDM	LM	Hybrid	yes	120	3.929	1965	0.000672892705

Table 4: Paired Gaussian-noise study. Noisy RMSE conditions on convergence; clean-reference RMSE uses every trial with a finite retained curve.

Noise	Model	Successes / 100	Mean noisy RMSE (A)	Mean clean-reference RMSE (A)
0.1%	SDM	100	0.000948868	0.000797286
0.1%	DDM	100	0.000865772	0.000737119
0.5%	SDM	100	0.002988591	0.001360800
0.5%	DDM	88	0.002915486	0.001365803
1.0%	SDM	100	0.005634783	0.002441317
1.0%	DDM	85	0.005462373	0.002485752
2.0%	SDM	100	0.011347343	0.004417775
2.0%	DDM	83	0.011274248	0.004685958

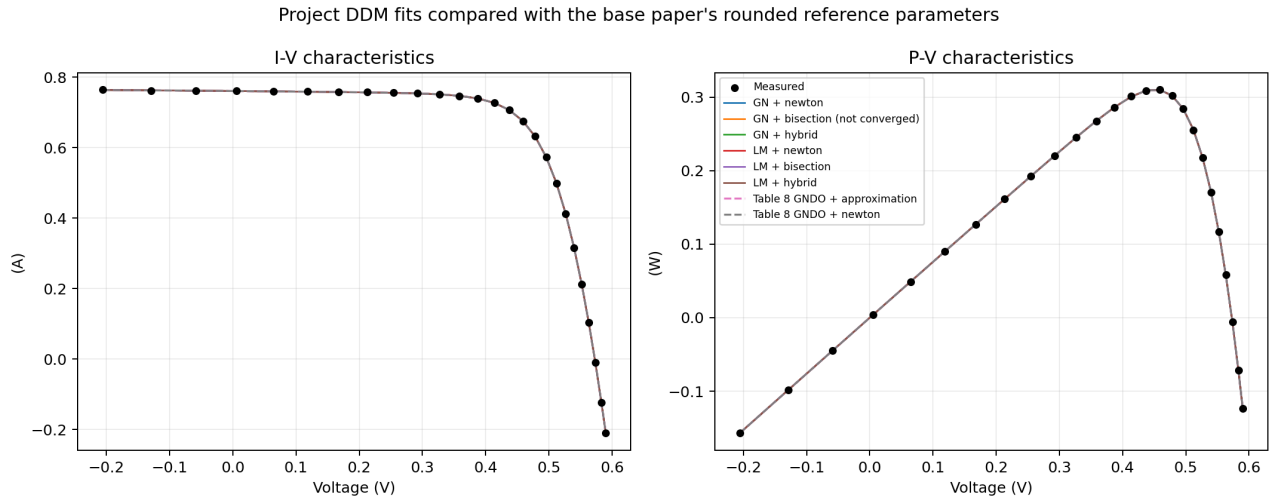


Figure 1: Measured RTC characteristics, deterministic SDM/DDM fits, and rounded base-paper references. The left panel is I-V and the right panel is P-V.

Project DDM fits compared with the base paper's rounded reference parameters

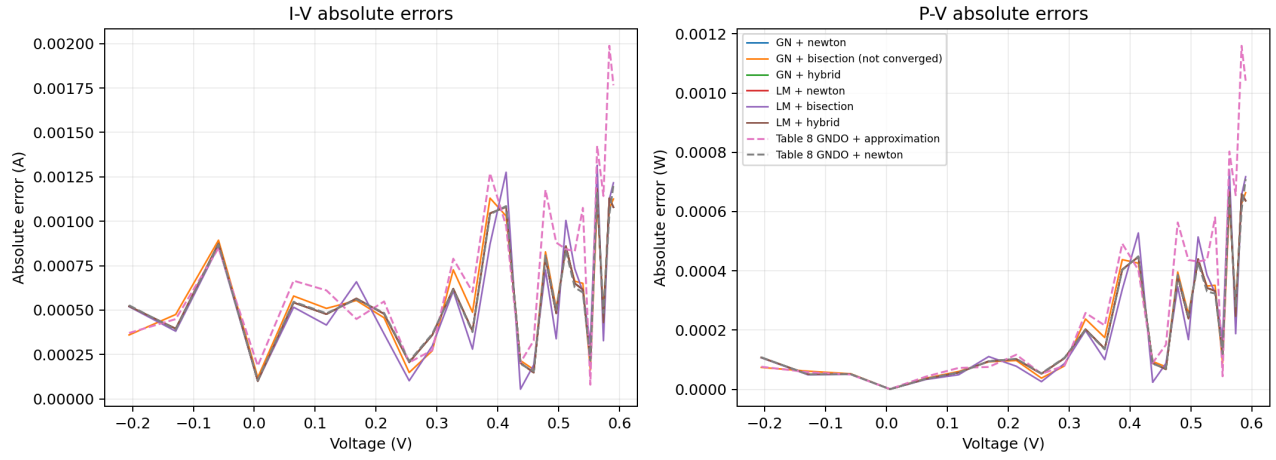


Figure 2: Absolute current error and voltage-weighted absolute power error for project fits and representative rounded-paper calculations.

Varied-initial-value ensemble: successes / total shown in legends; failed runs retained

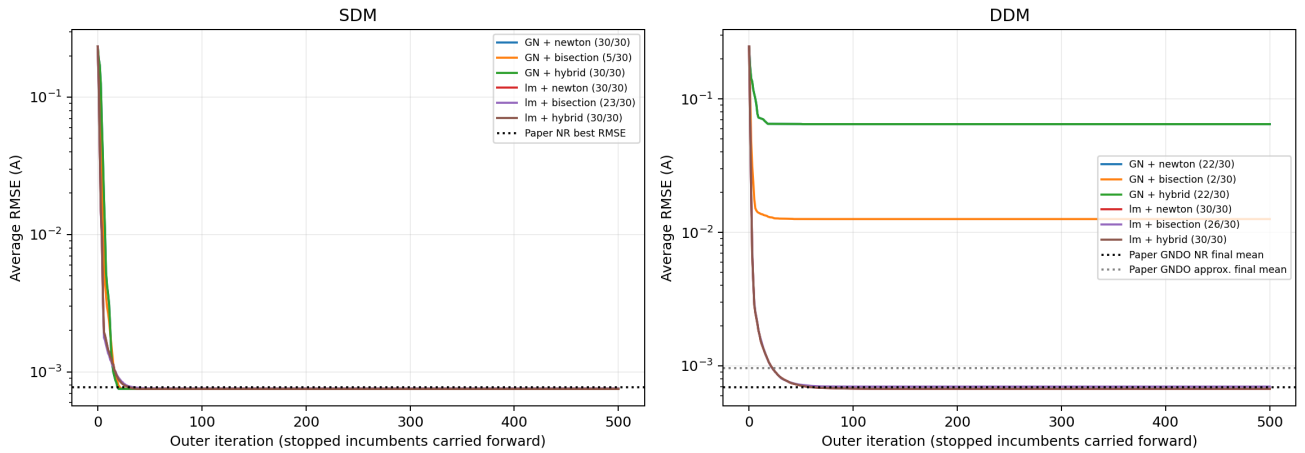


Figure 3: Mean RMSE over 30 varied starts. Failed finite runs remain in the average, and terminal paper values are reference lines rather than reconstructed histories.

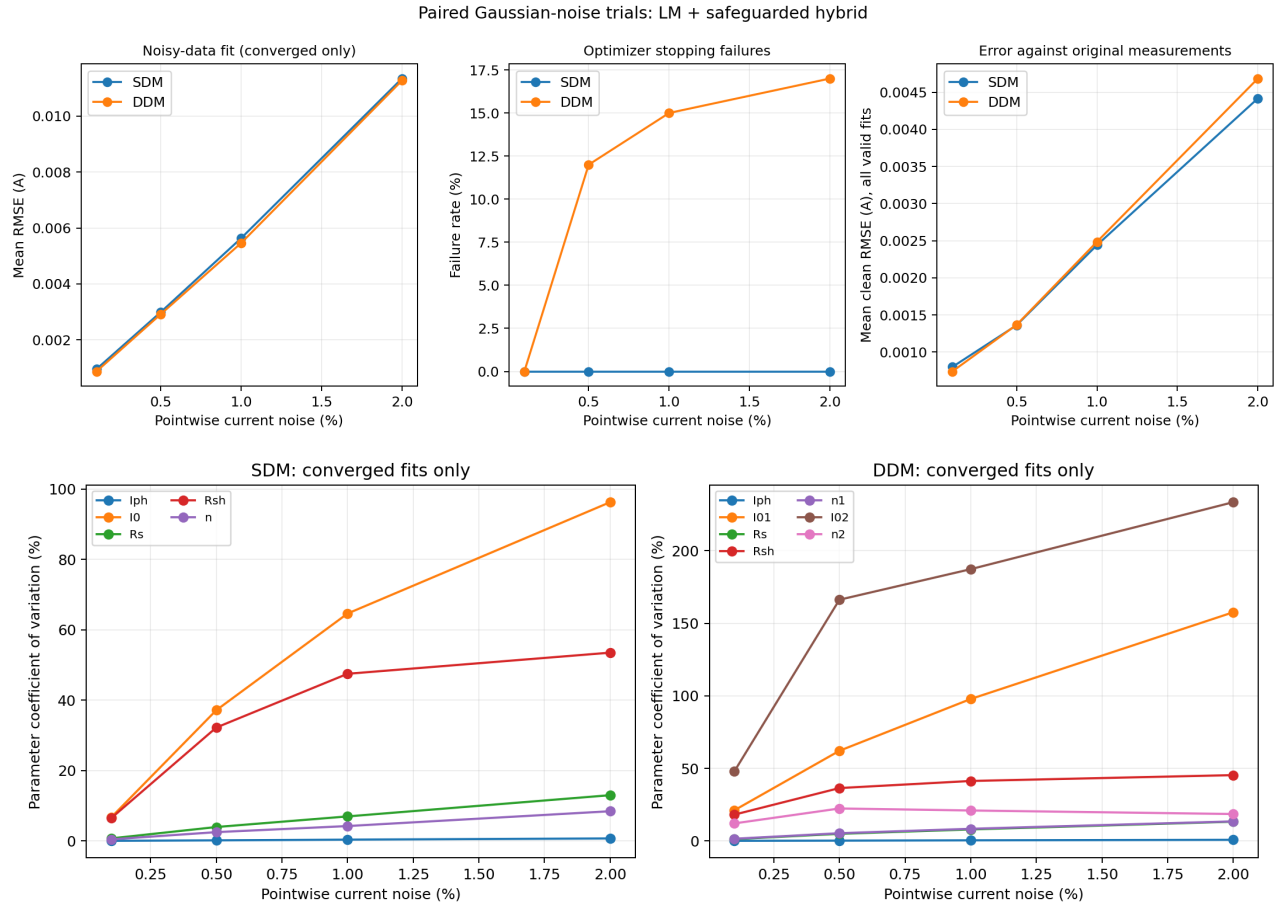


Figure 4: Paired noise analysis. Top: noisy-data fit, stopping failures, and error against the original measurements. Bottom: parameter coefficients of variation among converged fits; SDM and DDM use independent vertical ranges.

5 Conclusion and Future Work

The project implements and verifies a complete nested numerical pipeline for PV parameter extraction. Deterministic GN and LM reach clean RTC errors close to or below the base-paper reference levels, while the safeguarded hybrid provides a practical compromise between Newton speed and bisection protection. DDM lowers clean RMSE by 10.76%, but this modest curve improvement comes with much poorer conditioning, higher computational cost for LM, sensitivity to initial values, more noise-trial failures, and extreme variation in the two saturation currents. The main conclusion is therefore methodological: report convergence status, failure counts, conditioning, and parameter uncertainty in addition to RMSE.

The study is limited to the 26-point RTC cell, local seeded starts, finite parameter bounds, finite-difference derivatives, and a pointwise Gaussian current-noise model. The supplied PWP 201 data are not part of the reported experiment, and the five base-paper metaheuristics are reference results only. Future work should evaluate analytic or automatic-differentiation Ja-

cobians, trust-region and robust-loss variants, heteroscedastic voltage/current error, profile likelihood or Bayesian uncertainty, broader global initialization, and validation on PV modules and independent environmental conditions. A carefully matched work-count comparison with metaheuristic and Lambert W pipelines would further separate optimizer quality from current-solver cost.

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